Exercise sheet 5

Linear Optimization (311.160), WS 2026/27

Raphael Bleier 9 November 2026

Exercise 5.1

Maximize $x_1 + x_2$ subject to $2x_1 - x_2 \leq 16$, $-x_1 + x_2 \leq 5$, $x_1 + x_2 \leq 10$ and $x \geq 0$. 1. Determine the optimal solution using the simplex method. 2. Show that the obtained solution is not unique by performing a different pivot at the beginning. 3. Solve the problem graphically and formulate a general criterion for detecting when an optimal solution is not unique.

1. Slack form and the first simplex path

Introduce nonnegative slacks x_3, x_4, x_5:

$$\begin{aligned} 2x_1 - x_2 + x_3 &= 16 && \text{first inequality} \\ -x_1 + x_2 + x_4 &= 5 && \text{second inequality} \\ x_1 + x_2 + x_5 &= 10 && \text{third inequality} \\ z &= x_1 + x_2 && \text{objective} \end{aligned}$$

The initial basis is $B = (3, 4, 5)$, with nonbasis $N = (1, 2)$. Setting $x_1 = x_2 = 0$ gives

$$\begin{aligned} x_3 &= 16 && \text{first equation at } x_N = 0 \\ x_4 &= 5 && \text{second equation} \\ x_5 &= 10 && \text{third equation} \end{aligned}$$

All three values are nonnegative, so the basic solution is feasible. Solve the slacks for the basic variables:

$$\begin{aligned} x_3 &= 16 - 2x_1 + x_2 && \text{move } 2x_1 - x_2 \\ x_4 &= 5 + x_1 - x_2 && \text{move } -x_1 + x_2 \\ x_5 &= 10 - x_1 - x_2 && \text{move } x_1 + x_2 \\ z &= 0 + x_1 + x_2 && \text{objective, constant term is } 0 \end{aligned}$$

The reduced costs of (x_1, x_2) are $(1, 1)$. Both are positive. The largest value occurs twice. The tie is broken by the smaller original index, so x_1 enters. This is the same rule as `numpy.argmax` on the ordered nonbasis. In the dictionary $x_B = \bar{b} - \bar{A}x_N$, the column of x_1 is $(2, -1, 1)^T$. Only positive entries can drive a basic variable down to zero:

$$\begin{aligned} \frac{16}{2} &= 8 && \text{ratio for } x_3 \\ \frac{10}{1} &= 10 && \text{ratio for } x_5 \\ 8 &< 10 && \text{compare the two ratios} \end{aligned}$$

The coefficient of x_1 in the x_4 row is -1, so x_4 increases and does not limit the step. The leaving variable is x_3, with step length 8.

$$\begin{aligned} x_3 &= 16 - 2x_1 + x_2 && \text{dictionary row} \\ 2x_1 &= 16 - x_3 + x_2 && \text{add } 2x_1, \text{ subtract } x_3 \\ x_1 &= 8 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_2 && \text{divide by } 2 \end{aligned}$$

Substitute into the remaining rows:

$$
\begin{aligned}
x_4 &= 5 + x_1 - x_2 && \text{old row}\\
&= 5 + \left(8 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_2\right) - x_2 && \text{substitute } x_1\\
&= 5 + 8 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_2 - x_2 && \text{distribute}\\
&= 13 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_2 - x_2 && \text{add } 5+8\\
&= 13 - \tfrac{1}{2}x_3 - \tfrac{1}{2}x_2 && \tfrac{1}{2}x_2 - x_2 = -\tfrac{1}{2}x_2\\
x_5 &= 10 - x_1 - x_2 && \text{old row}\\
&= 10 - \left(8 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_2\right) - x_2 && \text{substitute } x_1\\
&= 10 - 8 + \tfrac{1}{2}x_3 - \tfrac{1}{2}x_2 - x_2 && \text{distribute the minus}\\
&= 2 + \tfrac{1}{2}x_3 - \tfrac{1}{2}x_2 - x_2 && \text{subtract}\\
&= 2 + \tfrac{1}{2}x_3 - \tfrac{3}{2}x_2 && \text{collect } x_2\\
z &= x_1 + x_2 && \text{objective}\\
&= 8 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_2 + x_2 && \text{substitute } x_1\\
&= 8 - \tfrac{1}{2}x_3 + \tfrac{3}{2}x_2 && \text{collect } x_2
\end{aligned}
$$

With $x_3 = x_2 = 0$ the new basic solution is $(x_1, x_2, x_3, x_4, x_5) = (8, 0, 0, 13, 2)$ and $z = 8$. The reduced cost of x_2 is $\frac{3}{2} > 0$, so the dictionary is not optimal. x_2 enters. The column of x_2 decreases x_4 at rate $\frac{1}{2}$ and x_5 at rate $\frac{3}{2}$. The coefficient in the x_1 row is $-\frac{1}{2}$, so x_1 increases.

$$
\begin{aligned}
\frac{13}{1/2} &= 26 && \text{ratio for } x_4\\
\frac{2}{3/2} &= \tfrac{4}{3} && \text{ratio for } x_5\\
\tfrac{4}{3} &< 26 && \text{compare the ratios}
\end{aligned}
$$

Thus x_5 leaves, with step length $\frac{4}{3}$.

$$
\begin{aligned}
x_5 &= 2 + \tfrac{1}{2}x_3 - \tfrac{3}{2}x_2 && \text{dictionary row}\\
\tfrac{3}{2}x_2 &= 2 + \tfrac{1}{2}x_3 - x_5 && \text{add } \tfrac{3}{2}x_2, \text{ subtract } x_5\\
x_2 &= \tfrac{4}{3} + \tfrac{1}{3}x_3 - \tfrac{2}{3}x_5 && \text{divide by } \tfrac{3}{2}
\end{aligned}
$$

$$
\begin{aligned}
x_1 &= 8 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_2 && \text{old row}\\
&= 8 - \tfrac{1}{2}x_3 + \tfrac{1}{2}\left(\tfrac{4}{3} + \tfrac{1}{3}x_3 - \tfrac{2}{3}x_5\right) && \text{substitute } x_2\\
&= 8 - \tfrac{1}{2}x_3 + \tfrac{2}{3} + \tfrac{1}{6}x_3 - \tfrac{1}{3}x_5 && \text{distribute } \tfrac{1}{2}\\
&= \tfrac{26}{3} - \tfrac{1}{2}x_3 + \tfrac{1}{6}x_3 - \tfrac{1}{3}x_5 && 8 + \tfrac{2}{3} = \tfrac{24}{3} + \tfrac{2}{3}\\
&= \tfrac{26}{3} - \tfrac{1}{3}x_3 - \tfrac{1}{3}x_5 && \tfrac{1}{2} - \tfrac{1}{6} = \tfrac{1}{3}\\
x_4 &= 13 - \tfrac{1}{2}x_3 - \tfrac{1}{2}x_2 && \text{old row}\\
&= 13 - \tfrac{1}{2}x_3 - \tfrac{1}{2}\left(\tfrac{4}{3} + \tfrac{1}{3}x_3 - \tfrac{2}{3}x_5\right) && \text{substitute } x_2\\
&= 13 - \tfrac{1}{2}x_3 - \tfrac{2}{3} - \tfrac{1}{6}x_3 + \tfrac{1}{3}x_5 && \text{distribute}\\
&= \tfrac{37}{3} - \tfrac{1}{2}x_3 - \tfrac{1}{6}x_3 + \tfrac{1}{3}x_5 && 13 - \tfrac{2}{3} = \tfrac{39}{3} - \tfrac{2}{3}\\
&= \tfrac{37}{3} - \tfrac{2}{3}x_3 + \tfrac{1}{3}x_5 && \tfrac{1}{2} + \tfrac{1}{6} = \tfrac{2}{3}\\
z &= 8 - \tfrac{1}{2}x_3 + \tfrac{3}{2}x_2 && \text{old objective}\\
&= 8 - \tfrac{1}{2}x_3 + \tfrac{3}{2}\left(\tfrac{4}{3} + \tfrac{1}{3}x_3 - \tfrac{2}{3}x_5\right) && \text{substitute } x_2\\
&= 8 - \tfrac{1}{2}x_3 + 2 + \tfrac{1}{2}x_3 - x_5 && \text{distribute } \tfrac{3}{2}\\
&= 10 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_3 - x_5 && \text{add } 8+2\\
&= 10 - x_5 && \text{the } x_3 \text{ terms cancel}
\end{aligned}
$$

Set the new nonbasic variables $x_3 = x_5 = 0$:

$$\begin{aligned} x_1 &= \tfrac{26}{3} && \text{constant term} \\ x_2 &= \tfrac{4}{3} && \text{constant term} \\ x_4 &= \tfrac{37}{3} && \text{constant term} \\ z &= 10 && \text{constant term} \end{aligned}$$

The reduced costs of (x_3, x_5) are $(0, -1)$. Neither is positive, so this basic feasible solution is optimal for this pivot sequence. Probe in the original inequalities, with $x_1 = \frac{26}{3}$ and $x_2 = \frac{4}{3}$:

$$\begin{aligned} 2x_1 - x_2 &= 2 \cdot \tfrac{26}{3} - \tfrac{4}{3} && \text{substitute} \\ &= \tfrac{52}{3} - \tfrac{4}{3} && \text{multiply} \\ &= \tfrac{48}{3} && \text{subtract} \\ &= 16 && \text{divide} \\ -x_1 + x_2 &= -\tfrac{26}{3} + \tfrac{4}{3} && \text{substitute} \\ &= -\tfrac{22}{3} && \text{add} \\ x_1 + x_2 &= \tfrac{26}{3} + \tfrac{4}{3} && \text{substitute} \\ &= \tfrac{30}{3} && \text{add} \\ &= 10 && \text{divide} \\ x_1 + x_2 &= 10 && \text{objective at this point} \end{aligned}$$

So $2x_1 - x_2 = 16 \le 16$, $-x_1 + x_2 = -\frac{22}{3} \le 5$ and $x_1 + x_2 = 10 \le 10$. Both coordinates are positive.

$$\boxed{x = \left(\tfrac{26}{3}, \tfrac{4}{3}\right)^T, \qquad z = 10.}$$

2. A different pivot at the beginning

Start again from

$$\begin{aligned} x_3 &= 16 - 2x_1 + x_2 && \text{initial dictionary} \\ x_4 &= 5 + x_1 - x_2 && \text{initial dictionary} \\ x_5 &= 10 - x_1 - x_2 && \text{initial dictionary} \\ z &= x_1 + x_2 && \text{objective} \end{aligned}$$

Now let x_2 enter, although its reduced cost equals that of x_1. The column of x_2 is $(-1, 1, 1)^T$. The first entry is negative, so x_3 increases.

$$\begin{aligned} \frac{5}{1} &= 5 && \text{ratio for } x_4 \\ \frac{10}{1} &= 10 && \text{ratio for } x_5 \\ 5 &< 10 && \text{compare the ratios} \end{aligned}$$

Hence x_4 leaves, with step length 5.

$$\begin{aligned} x_4 &= 5 + x_1 - x_2 && \text{dictionary row} \\ x_2 &= 5 + x_1 - x_4 && \text{add } x_2, \text{ subtract } x_4 \end{aligned}$$

$$\begin{aligned} x_3 &= 16 - 2x_1 + x_2 && \text{old row} \\ &= 16 - 2x_1 + (5 + x_1 - x_4) && \text{substitute } x_2 \end{aligned}$$

$$
\begin{aligned}
&= 16 - 2x_1 + 5 + x_1 - x_4 && \text{distribute} \\
&= 21 - 2x_1 + x_1 - x_4 && \text{add } 16 + 5 \\
&= 21 - x_1 - x_4 && \text{collect } x_1 \\
x_5 &= 10 - x_1 - x_2 && \text{old row} \\
&= 10 - x_1 - (5 + x_1 - x_4) && \text{substitute } x_2 \\
&= 10 - x_1 - 5 - x_1 + x_4 && \text{distribute the minus} \\
&= 5 - 2x_1 + x_4 && \text{collect terms} \\
z &= x_1 + x_2 && \text{objective} \\
&= x_1 + (5 + x_1 - x_4) && \text{substitute } x_2 \\
&= 5 + 2x_1 - x_4 && \text{collect } x_1
\end{aligned}
$$

With $x_1 = x_4 = 0$ we have $(x_1, x_2) = (0, 5)$, slacks $(21, 0, 5)$ and $z = 5$. The reduced cost of x_1 is $2 > 0$, so x_1 enters. Its column decreases x_3 at rate 1 and x_5 at rate 2. The coefficient in the x_2 row is -1, so x_2 increases.

$$
\begin{aligned}
\frac{21}{1} &= 21 && \text{ratio for } x_3 \\
\frac{5}{2} &= \tfrac{5}{2} && \text{ratio for } x_5 \\
\tfrac{5}{2} &< 21 && \text{compare the ratios}
\end{aligned}
$$

Thus x_5 leaves, with step length $\frac{5}{2}$.

$$
\begin{aligned}
x_5 &= 5 - 2x_1 + x_4 && \text{dictionary row} \\
2x_1 &= 5 + x_4 - x_5 && \text{add } 2x_1, \text{ subtract } x_5 \\
x_1 &= \tfrac{5}{2} + \tfrac{1}{2}x_4 - \tfrac{1}{2}x_5 && \text{divide by 2}
\end{aligned}
$$

$$
\begin{aligned}
x_2 &= 5 + x_1 - x_4 && \text{old row} \\
&= 5 + \left(\tfrac{5}{2} + \tfrac{1}{2}x_4 - \tfrac{1}{2}x_5\right) - x_4 && \text{substitute } x_1 \\
&= \tfrac{15}{2} + \tfrac{1}{2}x_4 - \tfrac{1}{2}x_5 - x_4 && \text{add } 5 + \tfrac{5}{2} \\
&= \tfrac{15}{2} - \tfrac{1}{2}x_4 - \tfrac{1}{2}x_5 && \tfrac{1}{2}x_4 - x_4 = -\tfrac{1}{2}x_4 \\
x_3 &= 21 - x_1 - x_4 && \text{old row} \\
&= 21 - \left(\tfrac{5}{2} + \tfrac{1}{2}x_4 - \tfrac{1}{2}x_5\right) - x_4 && \text{substitute } x_1 \\
&= \tfrac{37}{2} - \tfrac{1}{2}x_4 + \tfrac{1}{2}x_5 - x_4 && 21 - \tfrac{5}{2} = \tfrac{42}{2} - \tfrac{5}{2} \\
&= \tfrac{37}{2} - \tfrac{3}{2}x_4 + \tfrac{1}{2}x_5 && \text{collect } x_4 \\
z &= 5 + 2x_1 - x_4 && \text{old objective} \\
&= 5 + 2\left(\tfrac{5}{2} + \tfrac{1}{2}x_4 - \tfrac{1}{2}x_5\right) - x_4 && \text{substitute } x_1 \\
&= 5 + 5 + x_4 - x_5 - x_4 && \text{distribute} \\
&= 10 - x_5 && \text{the } x_4 \text{ terms cancel}
\end{aligned}
$$

Set $x_4 = x_5 = 0$:

$$
\begin{aligned}
x_1 &= \tfrac{5}{2} && \text{constant term} \\
x_2 &= \tfrac{15}{2} && \text{constant term} \\
x_3 &= \tfrac{37}{2} && \text{constant term} \\
z &= 10 && \text{constant term}
\end{aligned}
$$

The reduced costs of (x_5, x_4) are $(-1, 0)$. No reduced cost is positive, so this point is also optimal. It is not the point from part 1:

$$
\tfrac{26}{3} - \tfrac{5}{2} = \tfrac{52}{6} - \tfrac{15}{6} \qquad \text{common denominator}
$$

$= \frac{37}{6}$	subtract
$\frac{37}{6} \neq 0$	the first coordinates differ

Both objective values equal 10. Therefore the optimal solution obtained in part 1 is not unique. Probe with $x_1 = \frac{5}{2}$ and $x_2 = \frac{15}{2}$:

$2x_1 - x_2 = 2 \cdot \frac{5}{2} - \frac{15}{2}$	substitute
$= 5 - \frac{15}{2}$	multiply
$= -\frac{5}{2}$	subtract
$-x_1 + x_2 = -\frac{5}{2} + \frac{15}{2}$	substitute
$= \frac{10}{2}$	add
$= 5$	divide
$x_1 + x_2 = \frac{5}{2} + \frac{15}{2}$	substitute
$= 10$	add

So $2x_1 - x_2 = -\frac{5}{2} \leq 16$, $-x_1 + x_2 = 5 \leq 5$ and $x_1 + x_2 = 10 \leq 10$. Both coordinates are positive.

$$\boxed{x = \left(\tfrac{5}{2}, \tfrac{15}{2}\right)^T, \qquad z = 10, \quad \text{different from part 1.}}$$

3. Graphical solution and the uniqueness criterion

Rewrite the inequalities as bounds on x_2:

$x_2 \geq 2x_1 - 16$	first inequality, add x_2, subtract 16
$x_2 \leq x_1 + 5$	second inequality, add x_1
$x_2 \leq 10 - x_1$	third inequality, subtract x_1
$x_1 \geq 0$	sign restriction
$x_2 \geq 0$	sign restriction

The binding pairs that can produce a vertex are the axis intersections and the pairwise intersections of the three lines. Compute them one pair at a time.

$2x_1 = 16$	$x_2 = 0$ in the first equation
$x_1 = 8$	divide by 2

The point $(8, 0)$ satisfies $-8 \leq 5$ and $8 \leq 10$.

$x_2 = 5$	$x_1 = 0$ in the second equation

The point $(0, 5)$ satisfies $-5 \leq 16$ and $5 \leq 10$.

$x_1 + 5 = 10 - x_1$	second and third lines
$2x_1 = 5$	add x_1, subtract 5
$x_1 = \frac{5}{2}$	divide by 2
$x_2 = 10 - \frac{5}{2}$	third line
$= \frac{15}{2}$	subtract

Check the first inequality at this intersection:

$2 \cdot \frac{5}{2} - \frac{15}{2} = -\frac{5}{2}$	same calculation as the probe
$-\frac{5}{2} \leq 16$	compare with the bound

Next, the first and third lines:

$$
\begin{aligned}
2x_1 - (10 - x_1) &= 16 && \text{substitute } x_2 = 10 - x_1 \\
2x_1 - 10 + x_1 &= 16 && \text{distribute} \\
3x_1 &= 26 && \text{add } 10 \\
x_1 &= \tfrac{26}{3} && \text{divide by } 3 \\
x_2 &= 10 - \tfrac{26}{3} && \text{third line} \\
&= \tfrac{4}{3} && \text{subtract}
\end{aligned}
$$

Check the second inequality:

$$
\begin{aligned}
-\tfrac{26}{3} + \tfrac{4}{3} &= -\tfrac{22}{3} && \text{add} \\
-\tfrac{22}{3} &\le 5 && \text{compare with the bound}
\end{aligned}
$$

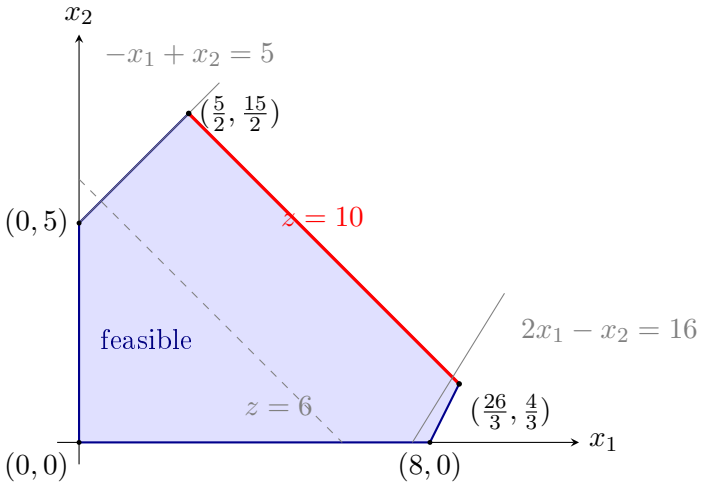
The remaining line pairs miss the feasible region. For the first and second lines,

$$
\begin{aligned}
2x_1 - (x_1 + 5) &= 16 && \text{substitute } x_2 = x_1 + 5 \\
x_1 - 5 &= 16 && \text{distribute} \\
x_1 &= 21 && \text{add } 5 \\
x_2 &= 26 && 21 + 5 \\
x_1 + x_2 &= 47 && \text{add} \\
47 &> 10 && \text{violates the third inequality}
\end{aligned}
$$

For the first line and $x_1 = 0$, the value $x_2 = -16$ is negative. The origin is the remaining feasible corner. The feasible vertices and their objective values are therefore

$$
\begin{aligned}
z(0,0) &= 0 && \text{sum of coordinates} \\
z(8,0) &= 8 && \text{sum of coordinates} \\
z(0,5) &= 5 && \text{sum of coordinates} \\
z\left(\tfrac{26}{3}, \tfrac{4}{3}\right) &= 10 && \text{sum, already checked} \\
z\left(\tfrac{5}{2}, \tfrac{15}{2}\right) &= 10 && \text{sum, already checked}
\end{aligned}
$$

The largest value is 10, attained at two vertices. A level line of the objective is $x_2 = -x_1 + t$. Its slope is -1, the same slope as $x_1 + x_2 = 10$. Shifting the line upward in the objective direction, the last contact with the feasible polygon is that whole edge, not a single vertex.



The two optimal basic solutions are the endpoints of this edge. The feasible region is convex, and the objective is linear. For $\lambda \in [0,1]$,

$$\begin{aligned} x(\lambda) &= \lambda\left(\tfrac{26}{3}, \tfrac{4}{3}\right)^T + (1-\lambda)\left(\tfrac{5}{2}, \tfrac{15}{2}\right)^T && \text{segment between the endpoints} \\ z\big(x(\lambda)\big) &= \lambda \cdot 10 + (1-\lambda)\cdot 10 && \text{both endpoints have objective 10} \\ &= 10 && \text{the } \lambda \text{ terms cancel} \end{aligned}$$

Every point of the segment is feasible and optimal. One explicit midpoint, $\lambda = \frac{1}{2}$, is

$$\begin{aligned} x_1 &= \tfrac{1}{2}\cdot\tfrac{26}{3} + \tfrac{1}{2}\cdot\tfrac{5}{2} && \text{substitute } \lambda \\ &= \tfrac{13}{3} + \tfrac{5}{4} && \text{multiply} \\ &= \tfrac{52}{12} + \tfrac{15}{12} && \text{common denominator} \\ &= \tfrac{67}{12} && \text{add} \\ x_2 &= \tfrac{1}{2}\cdot\tfrac{4}{3} + \tfrac{1}{2}\cdot\tfrac{15}{2} && \text{substitute } \lambda \\ &= \tfrac{2}{3} + \tfrac{15}{4} && \text{multiply} \\ &= \tfrac{8}{12} + \tfrac{45}{12} && \text{common denominator} \\ &= \tfrac{53}{12} && \text{add} \\ x_1 + x_2 &= \tfrac{67}{12} + \tfrac{53}{12} && \text{add the coordinates} \\ &= 10 && \text{sum of the numerators is 120} \end{aligned}$$

The midpoint also satisfies the other two inequalities:

$$\begin{aligned} 2x_1 - x_2 &= 2\cdot\tfrac{67}{12} - \tfrac{53}{12} && \text{substitute the midpoint} \\ &= \tfrac{134}{12} - \tfrac{53}{12} && \text{multiply} \\ &= \tfrac{81}{12} && \text{subtract} \\ &= \tfrac{27}{4} && \text{divide by 3} \\ -x_1 + x_2 &= -\tfrac{67}{12} + \tfrac{53}{12} && \text{substitute} \\ &= -\tfrac{14}{12} && \text{add} \\ &= -\tfrac{7}{6} && \text{divide by 2} \end{aligned}$$

So $2x_1 - x_2 = \frac{27}{4} \le 16$ and $-x_1 + x_2 = -\frac{7}{6} \le 5$. Criterion, for a maximization problem written as $z = \overline{z} + \overline{c}_N^T x_N$. If an optimal dictionary has $\overline{c}_j < 0$ for every nonbasic index j, then every feasible increase of a nonbasic variable decreases z. The optimal solution is unique. If some nonbasic reduced cost is zero and the minimum-ratio test for that variable returns a positive step, pivoting on it produces a different optimal basic solution. The optimum is then not unique, and every convex combination of optimal solutions is optimal. A zero reduced cost whose ratio is zero only changes the basis at the same point. That alone does not show a second solution. Graphically, in two variables, the optimal solution is not unique exactly when an optimal level line contains an edge of the feasible region. In this problem the level line $x_1 + x_2 = 10$ is the third constraint, restricted to the segment between the two basic solutions above.

$x(\lambda) = \lambda\left(\tfrac{26}{3}, \tfrac{4}{3}\right)^T + (1-\lambda)\left(\tfrac{5}{2}, \tfrac{15}{2}\right)^T,\ \lambda \in [0,1],\ z = 10,$
not unique iff an optimal dictionary has a zero reduced cost with positive step.

Exercise 5.2

Consider $\max c^T x$ *subject to* $Ax \le 0$ *and* $x \ge 0$. *Show that either* $x = 0$ *is optimal or the LP is unbounded.*

The zero vector is feasible:

$$A \cdot 0 = 0 \qquad \text{matrix times the zero vector}$$

$0 \le 0$	the inequality $Ax \le 0$ holds
$0 \ge 0$	the sign restriction holds
$c^T \cdot 0 = 0$	objective at $x = 0$

Either $c^T x \le 0$ for every feasible x, or there exists a feasible $\hat{x}$ with $c^T \hat{x} > 0$. Case 1. Assume $c^T x \le 0$ whenever $Ax \le 0$ and $x \ge 0$. Then

$c^T x \le 0$	assumption of this case
$c^T x \le c^T \cdot 0$	the right-hand side is the value at zero

for every feasible x. Therefore $x = 0$ is optimal. Case 2. Assume $\hat{x}$ is feasible and $c^T \hat{x} > 0$. For every $t \ge 0$ set $x(t) = t\hat{x}$. Then

$Ax(t) = t\,A\hat{x}$	factor t out of the matrix product
$t\,A\hat{x} \le 0$	$t \ge 0$ and $A\hat{x} \le 0$
$x(t) = t\hat{x}$	definition of the ray
$t\hat{x} \ge 0$	$t \ge 0$ and $\hat{x} \ge 0$

so $x(t)$ is feasible. Its objective value is

$c^T x(t) = t\,c^T \hat{x}$	factor t
$t\,c^T \hat{x} \to \infty$	$c^T \hat{x} > 0$ and $t \to \infty$

The feasible values are not bounded above, so the LP is unbounded. The two cases cover every c and every A. A positive cost coefficient by itself is not the second case: the corresponding unit step must still satisfy $Ax \le 0$.

Either $x = 0$ is optimal, or the LP is unbounded.

Exercise 5.3

Complete `choose_entering_variable_index` *and* `choose_leaving_variable_index` *from* `Exercise_05.py`. *Together with* `initial_simplex_dictionary` *from sheet 4, test them on Example 2.2. Submit* `LASTNAME_sheet_NUMBER.py` *and* `LASTNAME_sheet_NUMBER.pdf`.

Entering variable

The function reads `c_bar_N` and `N` from the dictionary of sheet 4. That dictionary stores $x_B = \overline{b} - \overline{A}x_N$ and $z = \overline{z} + \overline{c}_N^T x_N$, so a positive reduced cost is an improving direction for maximization. If no entry of `c_bar_N` exceeds 10^{-9}, the function returns `None`. Otherwise it returns the original index in `N` at which `c_bar_N` is maximal. A tie keeps the first maximum, which is the earlier position in `N`.

Leaving variable

Given the entering index, the function takes the matching column of `A_bar`. Rows whose entry does not exceed 10^{-9} are ignored, because those basic variables do not decrease. If no row remains, the function returns `None`: the entering variable can increase without a bound from the current dictionary, and the LP is unbounded along that ray. Otherwise it returns the original index in `B` of the smallest ratio $\overline{b}_i / \overline{A}_{i,\text{enter}}$. A tie keeps the first minimum in the current order of `B`.

Test on Example 2.2

Example 2.2 is $\max -x_1 + 2x_2 + 6x_3$ with

$$\begin{aligned} 2x_1 + 3x_2 + x_4 &= 10 && \text{first constraint of Example 2.2} \\ x_1 + 2x_3 + x_5 &= 8 && \text{second constraint} \\ x_1 + 2x_2 + x_3 + x_6 &= 6 && \text{third constraint} \end{aligned}$$

and $x \geq 0$. The initial call uses the zero-based basis `[3, 4, 5]`. Sheet 4 already gives

$$\bar{c}_N = (-1, 2, 6)^T, \qquad N = (0, 1, 2), \qquad B = (3, 4, 5).$$

The unique maximum reduced cost is 6, at position 2 of N:

$$\begin{aligned} 6 &> 2 && \text{compare the positive reduced costs} \\ 6 &> -1 && \text{compare with the remaining cost} \end{aligned}$$

so the entering index is 2, the variable x_3. Its column in $\overline{A}$ is $(0, 2, 1)^T$.

$$\begin{aligned} \frac{8}{2} &= 4 && \text{ratio in the } x_5 \text{ row} \\ \frac{6}{1} &= 6 && \text{ratio in the } x_6 \text{ row} \\ 4 &< 6 && \text{compare the ratios} \end{aligned}$$

The first row has coefficient 0 and is skipped. The leaving index is 4, the variable x_5. After this pivot the basic solution is $(0, 0, 4, 10, 0, 2)^T$ with value 24, as in Lecture 4. The completed file `Bleier_sheet_5.py` prints these two indices. `test_sheet5.py` also checks an optimal dictionary, an unbounded column, both tie rules and the second lecture pivot from this point to $z = 26$.

entering index 2 (x_3), leaving index 4 (x_5).